

Shih-Yun Tang 湯士昀

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Professional Appointment

- 2026 — current | Postdoctoral Fellow, The University of Texas at Austin, Austin, TX.
 - Supervisor: Professor Adam Kraus

Education

- 2023 — 2026 | Ph.D. in Physics and Astronomy, Rice University (transferred from NAU)
 - Advisors: Professor Christopher Johns-Krull (Rice) & Dr. Lisa Prato (Lowell Observatory)
- 2020 — 2022 | Ph.D., Astronomy and Planetary Science, Northern Arizona University (NAU)
 - Advisors: Dr. Lisa Prato (Lowell Observatory) & Professor Tyler Robinson (NAU)
- 2017 — 2019 | M.S., Physics, National Central University (NCU), Taiwan
Advisor: Professor Wen-Ping Chen
- 2013 — 2017 | B.S., Physics, National Central University (NCU), Taiwan

Research Interests

- Planet formation: Spectroscopic searches for the youngest hot Jupiters
- Stellar astrophysics: Star spots, star-disk interaction, and B field effect in young stellar objects
- Stellar cluster: Membership identification, cluster dynamics, and chemical component studies

Research Experience

- 2020 — Present | Graduate Research Assistant, Lowell Observatory
- July 2018 — September 2018 | Summer Intern, Max-Planck-Institute for Astronomy, Germany
 - Advisors: Dr. Bertrand Goldman & Dr. Chien-Cheng Lin
 - Topic: *Multi-dimensional parameterization of members in nearby star clusters: Gaia DR2 and photometric distancing*
- 2017 — 2019 | Graduate Research Assistant, NCU, Astronomy Institute
- June 2016 — August 2016 | Summer Intern @ The Chinese University of Hong Kong
 - Advisor: Prof. Li, Hua-bai
 - Topic: *Optical performance testing of the ASTE polarimeter for the ASTE telescope*
- 2015 — 2016 | Undergraduate Research Assistant, NCU, Astronomy Institute

Publications | 9/20 1st & 2nd author papers | h-index: 12 | See more on



First author citation count: 198+

1st & 2nd author papers:

- ☆ 2024: Tang, Shih-Yun; Johns-Krull, Christopher M.; Prato, L. et al., *Measuring the Spot Variability of T Tauri Stars Using Near-IR Atomic Fe and Molecular OH Lines*, [The Astrophysical Journal](#), 973:124
- ☆ 2023: Tang, Shih-Yun; Stahl, Asa G.; Prato, L. et al., *Star-Crossed Lovers DI Tau A and B: Orbit Characterization and Physical Properties Determination*, [The Astrophysical Journal](#), 950:92
- 2022: Pang, Xiao-Ying; Tang, Shih-Yun; Li, Yuqian et al., *3D Morphology of Open Clusters in the Solar Neighborhood with Gaia EDR3 II: Hierarchical Star Formation Revealed by Spatial and Kinematic Substructures*, [The Astrophysical Journal](#), 931:156
- ☆ 2021: Tang, Shih-Yun; Robinson, Tyler D.; Marley, Mark S. et al., *Impacts of Water Latent Heat on the Thermal Structure of Ultra-Cool Objects: Brown Dwarfs and Free-Floating Planets*, [The Astrophysical Journal](#), 922:26

- 2021: Stahl, Asa G.; **Tang, Shih-Yun**; Johns-Krull, Christopher M. et al., IGRINS RV: A Precision RV Pipeline for IGRINS Using Modified Forward-Modeling in the Near-Infrared, [The Astronomical Journal](#), 161:283
- ★ 2021: **Tang, Shih-Yun**; Stahl, Asa G.; Johns-Krull, Christopher M. et al., IGRINS RV: A Python package for precision radial velocities with Near-Infrared Spectra, [The Journal of Open Source Software](#), 6:62
- 2020: Zhang, Yu; **Tang, Shih-Yun**; Chen, W. P. et al., Diagnosing the Stellar Population and Tidal Structure of the Blanco1 Star Cluster, [The Astrophysical Journal](#), 889:99
- ★ 2019: **Tang, Shih-Yun**; Pang, Xiao-Ying; Yuan, Zhen et al., Disrupted Open Clusters with tidal tails: Coma Berenices and its neighboring group, [The Astrophysical Journal](#), 877:12
- ★ 2018: **Tang, Shih-Yun**; Chen, W. P.; Chiang, P. S. et al., Characterization of Stellar and Substellar Members in the Coma Berenices Star Cluster, [The Astrophysical Journal](#), 862:106

Co-author papers:

- 2025: Kutra, Taylor; Prato, Lisa; Tofflemire, Benjamin M, et al. (including **Tang, Shih-Yun**), Sites of Planet Formation in Binary Systems. II. Double the Disks in DF Tau, [The Astronomical Journal](#), 169:20
- 2024: Brooks, Hunter; Caselden, Dan; Kirkpatrick, J. Davy, et al. (including **Tang, Shih-Yun**), Discovery of 118 New Ultracool Dwarf Candidates Using Machine Learning Techniques, [The Astronomical Journal](#), 168:211
- 2024: Pang, Xiao-Ying; Liao, Siqi; Li, Jiadong, et al. (including **Tang, Shih-Yun**), The Present-Day Mass Function of Star Clusters in the Solar Neighborhood, [The Astronomical Journal](#), 966:169
- 2023: Pang, Xiao-Ying; Wang, Yifan; **Tang, Shih-Yun**, et al., Binary Star Evolution in Different Environments: Filamentary, Fractal, Halo and Tidal-tail Clusters, [The Astronomical Journal](#), 166:110
- 2022: Pang, Xiao-Ying; Li, Yuqian; **Tang, Shih-Yun**, et al., Dynamical Origin for the Collinder 132-Gulliver 21 Stream: A Mixture of three Co-Moving Populations with an Age Difference of 250 Myr, [The Astrophysical Journal Letters](#), 937:L7
- 2021: Mann, Andrew W.; Wood, Mackenna L.; Schmidt, Stephen P. et al. (including **Tang, Shih-Yun**), TESS Hunt for Young and Maturing Exoplanets (THYME) VI: an 11 Myr giant planet transiting a very low-mass star in Lower Centaurus Crux, [The Astronomical Journal](#), 163:156
- 2021: Lee, Yong-Hee; Johnstone, Doug; Lee, Jeong-Eun et al. (including **Tang, Shih-Yun**), The JCMT Transient Survey: Four Year Summary of Monitoring the Submillimeter Variability of Protostars, [The Astrophysical Journal](#), 920:119
- 2021: Li, Yezhang; Pang, Xiao-Ying; **Tang, Shih-Yun** et al., Evidence of Early-stage Tidal Structures of Open Clusters Revealed by Kinematics with Gaia EDR3, [Research Notes of the AAS](#), 5:173
- 2021: Pang, Xiao-Ying; Yu, Zeqiu; **Tang, Shih-Yun** et al., Disruption of Hierarchical Clustering in the Vela OB2 Complex and the Cluster Pair Collinder 135 and UBC7 with Gaia EDR3: Evidence of Supernova Quenching, [The Astrophysical Journal](#), 923:20
- 2021: Pang, Xiao-Ying; Li, Yuqian; Yu, Zeqiu; **Tang, Shih-Yun** et al., 3D Morphology of Open Clusters in the Solar Neighborhood with Gaia EDR 3: Its Relation to Cluster Dynamics, [The Astrophysical Journal](#), 912:162
- 2020: Pang, Xiao-Ying; Li, Yuqian; **Tang, Shih-Yun** et al., Different Fates of Young Star Clusters after Gas Expulsion, [The Astrophysical Journal Letters](#), 900:L4

Awards/Scholarship/Fellowship

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- 2025 William & Elva Gordon Fellowship
 - 2025 RSI Center for Planetary Origins to Habitability graduate fellowships (equiv. to about 30k)
 - 2024 NSBP-NSHP Beth Brown Memorial Award (Best Graduate Poster)
 - 2024 Rice Physics and Astronomy Department Graduate Student Travel Award
 - 2019 Best Presentation Award, Astronomical Society of the ROC (Taiwan) annual meeting
 - 2018 Undergraduate Excellent Thesis Award, The Physics Society of Taiwan
 - 2018 NCU scholarship for outstanding students
 - 2018 MPIA Summer Internship Fellowship (full support)
 - 2018 Best Poster Award, the Taiwan Physics Society annual meeting

- 2017 NCU, The ICT Solution Provider scholarship
- 2016 NCU, Fellowship for Exchange Student to Niigata University, Japan
- 2016 Best Poster Award, Astronomical Society of the ROC (Taiwan) annual meeting

Presentations

Invited Talks:

- 2024 Seminar, Lowell Observatory, Flagstaff (July 25)
- 2023 Colloquium, Institute of Astronomy, NCU, Taiwan (May 26)
- 2023 Seminar, Center of Astronomy and Gravitation, National Taiwan Normal University, Taiwan (May 25)

Contributed Talks & Posters

- ▶ 2025:
 - Star and Planet Formation in the Southwest Conference (December 9 - 11 @ Austin) — **Talk**
 - The Solar System in Context, NOIRLab Science conference (September 29 - October 2 @ Tucson) — **Talk**
 - The Astronomical Society of the Republic of China (Taiwan) annual meeting (May 16-18 @ Yunlin, Taiwan) — **Talk**
- ▶ 2024:
 - 2024 NSBP-NSHP Joint Conference (November 13-16 @ Houston, USA) — **Poster**
 - The 9th Emerging Researchers in Exoplanet Science Symposium (July 10-12 @ Cornell University, USA) — **Talk**
 - Cool Stars, Stellar Systems, and the Sun Conference (Cool Stars 22, June 24-28 @ San Diego University, USA) — **Poster**
- ▶ 2023:
 - The 8th Emerging Researchers in Exoplanet Science Symposium (June 19-20 @ Yale University, USA) — **Poster**
 - The 2023 Gordon Research Conference/Seminar on Origins of Solar Systems (June 10-16 @ South Hadley, USA) — **Poster**
 - The Astronomical Society of the Republic of China (Taiwan) annual meeting (May 19-21 @ Kaohsiung, Taiwan) — **Poster**
 - IGRINS YSO meeting (May 3-4 @ Austin, USA) — **Talk**
 - 241st AAS meeting (January 8-12 @ Seattle, USA) — **iPoster** [[iPoster link](#)]
- ▶ 2022:
 - 2022 Fall Flagstaff Astronomy Symposium (October 26 @ Flagstaff, USA) — **Talk**
 - 2022 Sagan Exoplanet Summer Hybrid Workshop: Exoplanet Science in the Gaia Era (July 25-29 @ Pasadena, USA) — **Poster**
 - Exoplanets IV (May 1-6 @ Las Vegas, USA) — **Poster**
 - 2022 Spring Flagstaff Astronomy Symposium (April 14 @ Flagstaff, USA) — **Talk**
- ▶ 2021:
 - The 6th Emerging Researchers in Exoplanet Science Symposium (May 24-26, online) — **Talk**
- ▶ 2020:
 - 2020 Flagstaff Astronomy Symposium (March 4 @ Flagstaff, USA) — **Talk**
- ▶ 2019:
 - Micro-Workshop for Stellar Initial Mass Function and Molecular Core Mass Function (November 2019 @ Taipei, Taiwan) — **Talk**

- The Astronomical Society of the Republic of China (Taiwan) annual meeting (May 17-19 @ Taichung, Taiwan) — **Talk**
- Star Cluster Workshop (April @ Kunming, China) — **Talk**
- ▶ 2018:
 - ESO Workshop: A revolution in stellar physics with Gaia and large surveys (September 2018 @ Warsaw, Poland) — **Poster**
 - Star Cluster Workshop (July @ Xinjiang, China) — **Talk**
 - The Astronomical Society of the Republic of China (Taiwan) annual meeting (May 18-20 @ Kinmen, Taiwan) — **Talk**
 - The Taiwan Physics Society Annual Meeting (June 24-26 @ Taipei, Taiwan) — **Poster**
- ▶ 2016:
 - The Astronomical Society of the Republic of China (Taiwan) annual meeting (May 13-15 @ Tainan, Taiwan) — **Poster**

Successful Proposals

- PI: McDonald Observatory 2.7m/IGRINS spectrographs, a total of **26 nights** from 2024 - 2025 (McD25-1-6, McD25-3-11)
- PI: McDonald Observatory 2.7m/Tull coudé spectrographs, a total of **38 nights** from 2022-2024 (McD22-3-8, McD23-1-7, McD23-3-5, and McD24-1-3)
- PI: Gemini North/GNIRS, Fast Turnaround, 1.5 hours (GN-2017B-FT-18)
- Co-I: McDonald Observatory 2.7m/Tull coudé spectrographs, 3 nights (McD23-2-5)
- Co-I: IRTF/iSHELL, 14.5 hours (2022B060)
- Co-I: CFHT/SPIRou, 19.3 hours (20BH33)

Open Source Projects | GitHub

- IGRINS RV: https://github.com/shihyuntang/igrins_rv
An open-source Python pipeline for extracting radial velocity (RV) for the Immersion GRating INfrared Spectrometer (IGRINS) instrument

Outreach & Media

- Featured interview, *Shih-Yun Tang: Decoding the Cosmos*, Rice University Wiess School of Natural Sciences (2025)
- Public talk, Astronomy on Tap: *Searching for the Youngest Hot Jupiter*, Flagstaff, AZ (2024)
- Article, *Solving the Planet Formation Riddle*, The Lowell Observer (quarterly newsletter), Issue 130, p. 10 (2023)
- Public program, *Meet an Astronomer*, Lowell Observatory (2022)

Teaching

- Guest lecture @ Dept. Astronomy and Planetary Science, NAU, **Spring 2023**
 AST 210 — *Celebrating Diversity in Astronomy*
- Guest lecture @ Dept. Astronomy and Planetary Science, NAU, **Fall 2022**
 AST 391 — *Astrophysics: stars*
- Teaching Assistant @ Dept. Astronomy and Planetary Science, NAU, **Fall 2022**
 AST 401L — *Observational Astronomy Laboratory*
- Lab Instructor @ Dept. Astronomy and Planetary Science, NAU, **Fall 2022**
 AST 190L — *The Planets Laboratory*
- Leading hands-on sessions helper @ Sagan Exoplanet Summer Workshop, Pasadena, USA: Exoplanet Science in the Gaia Era, **Summer 2022**
- Guest lectures @ Dept. Astronomy and Planetary Science, NAU, **Fall 2021**
 AST 391 — *Astrophysics: stars*

- Co-instructor (with Christian J. Tai Udovicic) on programming course
@ Dept. Astronomy and Planetary Science, NAU, **Fall 2021**
Course website: https://cjt.u.github.io/spirl/f21_about.html
- Leading hands-on sessions helper @ Sagan Exoplanet Summer Virtual Workshop (online):
Circumstellar Disks and Young Planets, **Summer 2021**
- Teaching Assistant @ Department of Physics, NCU, **Spring 2018**
PH 1024 — General Physics Laboratory
- Teaching Assistant @ Department of Physics, NCU, **Fall 2018**
PH 1023 — General Physics Laboratory

Student Project Mentor/Supervision

- Co-supervised (with Dr. Xiaoying Pang) undergrad final year project on data visualization using Plotly Dash App — Jiayu Li, XJTLU, **Fall 2021 — Spring 2022**
Project web page: <http://3doc-morphology.lowell.edu>
- Mentoring undergraduate research — Hunter Brooks, NAU, **Fall 2021**. Project: *Discovery of New Low-Mass Objects and Brown Dwarfs in Blanco 1's Tidal Tails*